POINTED SIMPLY CONNECTED 3-TYPES ARE CLASSIFIED BY TRIPLES (A, B, k)

Purpose. Informal proof for `doc/postnikov-classification-handoff.txt`: an equivalence of types

$$\mathsf{SC3}_\ell \;\simeq\; \mathsf{Triples}_\ell$$

whose inverse is the realisation $(A, B, k) \mapsto (\mathrm{Fib}k, \ldots)$ of `PostnikovRealization.Triple`. The cocycle k is an actual pointed map; nothing is truncated or quotiented.

Attribution. The proof is the equivalence between $(n{+}1)$-groups and untruncated Sính triples of Perry Hart and Owen Milner, *Classifying 2-Groups in Homotopy Type Theory*, LICS 2026, LIPIcs vol. 380, DOI `10.4230/LIPIcs.LICS.2026.55`, §§8–9 (Lemma 40, Theorem 41, Note 42), specialised to $n = 2$ and to simply connected types. The paper states Theorem 41 as a result of the paper; the repository's `README.md` and `Sinh/README.md` call it "Owen Milner's type equivalence", and we follow that attribution. The delooping and Eilenberg–MacLane machinery of Unit (`sag/em`) is David Wärn's (DW1); the sphere universal-property strategy of the existing development is Wärn's, and plays no role here.

1. Sources read

Paper: `CANONICITY/2groups.pdf` (25 pp., LICS 2026 version), §8 (Definition 35, Theorem 36, Example 37, Theorem 38, Corollary 39) and §9 (Lemma 40, Theorem 41, Note 42, Theorem 43).
Agda: `https://github.com/PHart3/2-groups-agda`, branch `lics`, commit `9ec62dd454a8330278fdfce5cac9d1a` (9 July 2026), cloned read-only (shallow) into the session scratchpad; not built. Files read in full: `Sinh/Sinh-classif.agda`, `Sinh/Sinh-action.agda`, `Sinh/Sinh-cocycle.agda`, `Final/Final-thms.agda`, `HoTT-Agda/core/lib/types/N-groups.agda`, `HoTT-Agda/core/lib/types/Torsor.agda`, `HoTT-Agda/theorems/` (first 70 lines), `HoTT-Agda/theorems/homotopy/EM-unique.agda`, `HoTT-Agda/theorems/homotopy/EM-Grp-equiv` (the equivalence `EM-Ab-equiv`); the statements of `orthog-conn-trunc` (`lib/NConnected.agda`), `AbGrp-isGrpd` (`lib/groups/Isomorphism.agda`), `Π⊙-==-ext` (`lib/types/PtdFibration.agda`), `⊙EM-Torsors-≃` (`homotopy/EilenbergMacLane.agda`). Not read: the proofs of `EM-Torsors-≃`'s dependencies, `Ω^'S-abgroup`, the EM construction. Postulates: the repository's own `findpostulates` script lists only HIT and univalence postulates; a full `grep postulate` finds, besides lines tagged HIT or univalence, one untagged block, `EilenbergMacLane1/Core.agda` line 44 (`emloop-coh'`, `EM₁-level'`: the coherence and truncation constructors of the $K(G, 1)$ HIT) and `Equivalence2.agda` line 223 ("rest of univalence axiom"). The Sính equivalence uses univalence, function extensionality, truncation and EM HITs; no choice.

Correspondence.

paper	Agda	statement / Unit counterpart
Def. 35, Thm 36(1)	`Torsors`, `tors-ptd-is-trivial` (`torsors/Delooping`)	$TX = \sum_Y \lVert Y \rVert \times \prod_y X \simeq_* (Y, y)$; $(\tau = \tau_c) \simeq \mathrm{pr}_1\tau$ when T_*X is contractible. Unit: `Deloop.Torsors` (small: Y ranges over $\mathrm{BAut}\,X$), `Deloop.Torsors.Deloop.F.tr/equiv` ($(\mathsf{ptT} = t) \simeq \mathsf{el}\,t$; direction from the centre).

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Ex. 37	⊙EM-Torsors-≃	$K(A,n{+}1) \simeq_* TK(A,n)$. Unit: *by definition*, $K(B,4) = \mathsf{Tor}\,K(B,3)$ (`EM.Step`, `Presentation.Pres.K4-def`).
Thm 38	`EM-Ab-equiv` (`EM-Grp-equiv`)	$K(-,n) : \mathrm{Ab} \simeq \mathcal{U}^*_{\geq n-1,\leq n}$, inverse π_n. Unit: **new** (Thm. 8) at $n = 2,3$, from `Presentation.Rec`, a new `Rec2`, the maps `KK.e`, and a new path lemma for `Ab`.
Cor. 39	`EM-Ω-==-ext` (`EM-unique`)	$(K(A,n) \simeq_* X) \simeq \mathrm{conn} \times \mathrm{trunc} \times (\pi_n X = A)$. Unit: **not needed** — we work over $\mathsf{Stage}[1]$ instead of Ab (Remark 11).
Lemma 40	`N-Grps-≃` (`lib/types/N-groups`): `to`, `from`, `to-from` (via `NG-v2-to-==`), `from-to`	$\mathcal{U}^*_{\geq 0,\leq n} \simeq \sum_B \sum_{F:B\to\mathcal{U}_{\geq n-1,\leq n}} F(\mathrm{pt})$. Unit: **new** at the simply connected level (Prop. 12).
Thm 41	`NGrp-Sinh-≃-pre`, `NGrp-Sinh-≃`: `orthog-contr`, `rearrange`, `sub-rearrange`, `EM-Ω-==-ext`, `tors-ptd-is-trivial`, `Π⊙-==-ext`	the chain of equivalences. Unit: **new** (Prop. 9), simplified: the module is constant (simply connected base), the contraction is Lemma 5.
(in Thm 41)	`orthog-conn-trunc`, `AbGrp-isGrpd`	fillers $X \to \mathrm{Ab}$ over 1 are contractible for n-connected X, Ab a 1-type. Unit: Lemma 5 with $\mathsf{Stage}[1]$ (a 1-type, Lemma 4) in place of Ab.
Note 42	`Sinh-cocycle`: `NGrp-Sinh-coc-fib-≃`	$X \simeq \sum_u (\kappa(u) = b_u)$. Unit: the agreement with the realisation (Prop. 13); same proof (Theorem 36(1) fibrewise).
Thm 43	`Sinh-classif-set`	set-truncated version. Unit: **out of scope** (not wanted).

2. Statement

Unit indices are shifted by two: $\mathsf{isConn}\,(m{+}2)$ is m-connected, $\mathsf{isTrunc}\,(m{+}2)$ is m-truncated. Throughout, ℓ is a level, $\mathcal{U} = \mathcal{U}_\ell$, and $K(A,m) = \texttt{KK[A].K}\,m$ with base point $\texttt{KK[A].pt}\,m$.

Definition 1.

$$\mathsf{SC3}_\ell = \textstyle\sum_{X:\mathcal{U}_\ell} \sum_{x:X} \mathsf{isConn}\,3\,X \times \mathsf{isTrunc}\,5\,X \;:\; \mathcal{U}_{\ell+1}, \qquad \mathsf{Triples}_\ell = \textstyle\sum_{A:\mathrm{Ab}_\ell} \sum_{B:\mathrm{Ab}_\ell} K(A,2) \to_* K(B,4) \;:\; \mathcal{U}_{\ell+1}.$$

Ab_ℓ is the record `Torsor.Ab` (carrier in $\mathcal{U}_\ell$), a type in $\mathcal{U}_{\ell+1}$. For $n = 0,1$, $\mathsf{Stage}[n]$ is the record `EM.Stage[n]`: a type $X : \mathcal{U}_\ell$, a point, $\mathsf{isConn}\,(n{+}3)\,X$, $\mathsf{isTrunc}\,(n{+}4)\,X$ (pointed $(n{+}1)$-connected $(n{+}2)$-types), in $\mathcal{U}_{\ell+1}$.

The realisation is $\rho(A,B,k) = (\mathrm{Fib}k, (\mathrm{pt}, k_0), \texttt{conn3}, \texttt{trunc5})$ with $\mathrm{Fib}k = \sum_{x:K(A,2)} k(x) = \mathrm{pt}$ (`PostnikovRealization.Triple`).

Theorem 2. *$\rho : \mathsf{Triples}_\ell \to \mathsf{SC3}_\ell$ is an equivalence (in $\mathcal{U}_{\ell+1}$).*

The proof is a chain of equivalences

$$\mathsf{Triples} \overset{\text{Thm 8}}{\simeq} \textstyle\sum_{V:\mathsf{Stage}[0]} \sum_{W:\mathsf{Stage}[1]} V \to_* \mathsf{Tor}\,W \overset{\text{Prop 9}}{\simeq} \sum_{V:\mathsf{Stage}[0]} \mathsf{Fam}\,V \overset{\text{Prop 12}}{\simeq} \mathsf{SC3}$$

whose composite Φ is pointwise equal to ρ (Prop. 13); hence ρ is an equivalence (`Equivalences.homotopy` at level $\ell{+}1$).

3. Generic tools

Lemma 3 (paths of pointed types with a property)**.** *For a family of propositions $P : \mathcal{U} \to \mathcal{U}$, the type $\sum_X \sum_{x:X} P\,X$ has $((X,x,p) = (Y,y,q)) \simeq (X,x) \simeq_* (Y,y)$, with* idPt *over* refl*. In particular a pointed equivalence gives a path. Applied to* SC3 *(with $P =$ isConn 3 $\times$ isTrunc 5) and, through record η (proved by the eliminator* `split`*), to* Stage[n].

Proof. The total space $\sum_{(Y,y,q)}(X,x) \simeq_* (Y,y)$ is a retract of $\sum_{(Y,y,e)} P\,Y$ over the contractible `PtUniv` (`PtEquivOps.pt-univ`), whose fibre at the centre is the proposition $P\,X$, inhabited by p; so it is contractible; the fundamental theorem of identity types (`FundThm.FT`) gives the equivalence. □

Lemma 4. Stage[1] *is a 1-type: by Lemma 3 (through the record) its identity types are equivalent to $(X,x) \simeq_* (Y,y)$ between 2-connected 3-types, sets by* `Lemma4.cor5-equiv` *($t = 1$, $c = 3$), read at level $\ell{+}1$ by* `TruncUp.trunc-up`*.*

Lemma 5 (constant maps)**.** *Let D be 1-connected (*isConn 3*), Z a 1-type (*isTrunc 3*), $g : D \to Z$. Then $\sum_{z:Z} \prod_{d:D} z = g(d)$ is contractible.*

Proof. Contractibility is a proposition and D is merely inhabited; take d_0. The type is a retract of the pointed sections $\sum_{h:\prod_d gd_0 = gd} h(d_0) =$ refl: forth $(z,h) \mapsto (h',$ trans-sym-l) with $h'(d) = h(d_0)^{-1} \cdot h(d)$; back $h \mapsto (gd_0, h)$. The retraction law $(gd_0, h') = (z,h)$ is proved for an arbitrary $p : z = gd_0$ in place of $h(d_0)$, by J on p (right end fixed, `idJR`); at $p =$ refl it is $\mathsf{refl}^{-1} \cdot h(d) = h(d)$ pointwise, by function extensionality. The pointed sections are contractible by `Lemma4.lemma4` ($t = 0$, $c = 2$: a family of sets over a 1-connected base), at level $\ell{+}1$ since Z is large; its connectedness hypothesis is transported from level ℓ by `conn-lift` (induction on the index, comparing the two propositional truncations with the level-polymorphic eliminator; no resizing). This is Hart–Milner's `orthog-conn-trunc` at the level used. □

Lemma 6 (connectedness of a total space)**.** *If $f : A \to B$ has* isConn m *fibres and B is* isConn m*, then A is* isConn m*.*

Proof. Induction on m, dual to `ConnectedMaps.conn-up` (which goes from domain to codomain): $\|A\|$ from $\|B\|$ and a fibre; for a, a', ap$f$: $(a = a') \to (fa = fa')$ has isConn $(m{-}1)$ fibres (`ConnectedMaps.fib-ap-conn`) and $fa = fa'$ is isConn $(m{-}1)$. □

For $F : V \to \mathcal{U}$, $\mathrm{pr}_1 : \sum_v Fv \to V$ has fibres Fv (up to the equivalence $\mathrm{fib}_{\mathrm{pr}_1}(v) \simeq Fv$), so V and all Fv in isConn 3 give $\sum F$ in isConn 3.

Lemma 7 (paths in Ab)**.** *An additive bijection $h : G.A \simeq H.A$ gives $G = H$ in* Ab*.*

Proof. Induction on the equivalence (`Torsor.eqind`) reduces to $h = \mathrm{id}$, with G and H on the same carrier and $G.\mathrm{add} = H.\mathrm{add}$ pointwise. Then o agree (`EMMaps.hom-unit` of the identity), inv agree (inverses are unique), add agree by function extensionality, and the remaining fields are propositions (setA by `Torsor.isProp-isSet`, laws in a set). Record η by `split` on both sides. □

4. Theorem 38 at $n = 2, 3$

Let $\pi_3(W) =$ `Presentation.O3[W.X,W.pt,W.trunc].ab` (Ω^3 with $\cdot$, commutative by `EH`) and $\pi_2(V) =$ `O2ab`, the same at $\Omega^2 V$ (EH at V). Let $K_2(A) =$ `EMn[A].Kstage` 0 and $K_3(B) =$ `EMn[B].Kstage` 1; by definition $K_2(A).X = K(A,2)$ and $K_3(B).X = K(B,3)$, and Tor $(K_3 B) =$ `Step[1,Kstage 1].next`$.X$ is, by definition, $K(B,4)$ with its base point.

Theorem 8. *$K_2 : \mathrm{Ab} \to \mathsf{Stage}[0]$ and $K_3 : \mathrm{Ab} \to \mathsf{Stage}[1]$ are equivalences, with inverses π_2, π_3.*

Proof. $K_n(\pi_n W) = W$: Lemma 3 applied to the recognition equivalence $K(\pi_n W, n) \simeq_* W$, which is `Presentation.Rec.e3` for $n = 3$ and a new `Rec2.G2` for $n = 2$ (the steps `G1`, `G2` of `Rec`, with ΩF replaced by V; Cor. 12 then Cor. 11 with $c = 1$, Lemma 5 = `deloop-equiv`). $\pi_n(K_n B) = B$: Lemma 7 applied to the composite

$$\Omega^3 K(B,3) \xrightarrow{\Omega^2 e_2} \Omega^2 K(B,2) \xrightarrow{\Omega e_1} \Omega K(B,1) \xrightarrow{e_0} B$$

($e_m =$ `KK.e` m; for $n = 2$ the last two), an equivalence (looping of pointed equivalences, `LoopIso.loopPE`) and additive: the addition of $\pi_3 =$ `O3.ab` is concatenation in $\Omega(\Omega^2 K(B,3))$, the outermost loop, which $\Omega(\Omega e_2)$ preserves by `PiOne.Induced.loop-mult`; then Ωe_1 the same, and e_0 is the band, a homomorphism (`EMSpace.loop0-hom`). □

Only the existence of the two paths is used (bi-invertibility), not their values; the 1-truncatedness of Ab (Hart–Milner's `AbGrp-isGrpd`) is *not* used.

5. The fibrewise step (Theorem 41 at a simply connected base)

Fix $V : \mathsf{Stage}[0]$, $X = V.X$, $v_0 = V.\mathrm{pt}$. For $W : \mathsf{Stage}[1]$ let $\mathsf{Tor}\, W = \sum_{c:\mathrm{BAut}} \|\mathsf{el} c\| \times \mathsf{Mu}_W(\mathsf{el} c)$ with $\mathsf{Mu}_W\, Y = \prod_{y:Y} (W.X, W.\mathrm{pt}) \simeq_* (Y, y)$ and centre ptT_W (`Deloop.Torsors`, `EM.Step[1,W]`). Note the direction: Unit's tr_t goes from the centre, $(\mathsf{ptT} = t) \simeq \mathsf{el}\, t.1$, while the paper's Theorem 36(1) reads $(\tau = \tau_c) \simeq \mathrm{pr}_1 \tau$; Prop. 13 depends on this. Let

$$\mathsf{C}\, Y = \mathsf{isConn}\, 4\, Y \times \mathsf{isTrunc}\, 5\, Y, \qquad \mathsf{Fam}\, V = \textstyle\sum_{F:X\to\mathcal{U}} F(v_0) \times \prod_v \mathsf{C}(Fv) \;:\; \mathcal{U}_{\ell+1}.$$

Proposition 9. *$\sum_{W:\mathsf{Stage}[1]} (X, v_0) \to_* (\mathsf{Tor} W, \mathsf{ptT}_W) \;\simeq\; \mathsf{Fam}\, V$, by $(W, k, p) \mapsto (v \mapsto \mathsf{el}(kv).1,\ \mathsf{tr}_{kv_0}(p^{-1}),\ c)$, where $\mathsf{tr}_t : (\mathsf{ptT} = t) \simeq \mathsf{el}\, t.1$ is Theorem 36(1) (`F.tr`, transport of the centre's point) and c is computed from the structure (step 4 below).*

Proof. A chain of equivalences.

1. $(X, v_0) \to_* \mathsf{Tor} W = \sum_{k:X\to\mathsf{Tor}W} kv_0 = \mathsf{ptT} \simeq \sum_k \mathsf{el}(kv_0).1$, fibrewise by $p \mapsto \mathsf{tr}_{kv_0}(p^{-1})$ (`sym-equiv` then `F.equiv`).
2. $\mathsf{Tor} W \simeq \mathsf{Tor}^L W := \sum_{Y:\mathcal{U}} \|Y\| \times \mathsf{Mu}_W Y$, $(c, h, m) \mapsto (\mathsf{el} c, h, m)$, with inverse $(Y, h, m) \mapsto (c_Y,\ (h, m)$ transported along $p_Y^{-1})$, where $(c_Y, p_Y : \mathsf{el} c_Y = Y)$ is the centre of $\mathsf{Fib}\, Y = \sum_c \mathsf{el} c = Y$, contractible by `Torsors.fib-contr` from a point of Y; since $\mathsf{Fib}\, Y$ is a proposition (`fib-prop`: el is an embedding) $\|Y\|$ suffices (the text of `Presentation.Pres.tor-fib`, re-proved outside `Pres`). The two inverse laws: J on p_Y; and $\mathsf{Fib}(\mathsf{el} c)$ being a proposition, $(c_{\mathsf{el} c}, p) = (c, \mathsf{refl})$. Postcomposition gives $(X \to \mathsf{Tor} W) \simeq (X \to \mathsf{Tor}^L W)$, and `SphereAxiom.sigma-base` (a parameter-free lemma of that file, at level $\ell{+}1$; no sphere axiom) transports the family $k' \mapsto (k' v_0).1$, which on the image of k is $\mathsf{el}(kv_0).1$ by definition.
3. $\sum_{k':X\to\mathsf{Tor}^L W} (k' v_0).1 \;\simeq\; \sum_{F:X\to\mathcal{U}} \sum_{m:\prod_v \|Fv\|\times\mathsf{Mu}_W(Fv)} Fv_0$, by definition (regrouping; both inverse laws refl).
4. Summing over W and regrouping: $\sum_W (\ldots) \simeq \sum_F F(v_0) \times S(F)$ with $S(F) = \sum_{W:\mathsf{Stage}[1]} \prod_v \|Fv\| \times \mathsf{Mu}_W(Fv)$. Lemma 10: $S(F) \simeq \prod_v \mathsf{C}(Fv)$.

□

Lemma 10 (key lemma)**.** *For $F : X \to \mathcal{U}$ (X pointed and 1-connected), $S(F)$ and $\prod_v \mathsf{C}(Fv)$ are propositions and logically equivalent; hence equivalent (`prop-equiv`).*

Proof. (levels, $S \Rightarrow \mathsf{C}$) Given (W, m) and v: $\mathsf{C}(Fv)$ is a proposition and $\|Fv\|$ holds; for $y : Fv$, $m\, v\, y : W.X \simeq Fv$ transfers $\mathsf{isConn}\, 4$ and $\mathsf{isTrunc}\, 5$ from W (`to-C`).

($S(F)$ is a proposition) It suffices that $S(F)$ is contractible when inhabited (`prop-of-contr`). From (W, m) we have $c = \texttt{to-C}(W, m)$, hence $G : E \to \mathsf{Stage}[1]$, $G(v, y) = (Fv, y, c\,v)$ with $E = \sum_v Fv$. By Lemma 3 ($\mathsf{Stage}[1]$ form), $W \simeq_* G(e)$ is equivalent to $W = G(e)$, so $S(F)$ is a retract of $\sum_W \prod_{e:E} W = G(e)$ (the $\|Fv\|$ components are propositions, filled by c; the retraction law is `sec-eq` of the equivalence, pointwise). The latter is contractible by Lemma 5: E is 1-connected (Lemma 6, using c) and $\mathsf{Stage}[1]$ is a 1-type (Lemma 4).

($\mathsf{C} \Rightarrow S$) Given c: E is $\mathsf{isConn}\,3$; Fv_0 is merely inhabited and $S(F)$ is a proposition (above), so we may take $y_0 : Fv_0$. Put $W_0 = (Fv_0, y_0, c\,v_0)$. The family $Q(v, y) = (Fv_0, y_0) \simeq_* (Fv, y)$ over E consists of sets (`cor5-equiv`), so by `lemma4` ($t = 0$, $c = 2$) the pointed sections of Q through idPt at (v_0, y_0) are contractible; the centre gives $\mathsf{Mu}_{W_0}(Fv)$ for all v (the text of `Presentation.Pres.ext`, re-proved outside `Pres`). □

Remark 11 (route). Hart–Milner keep a family $M : X \to \mathrm{Ab}$ and convert $W \simeq_* (Fv, y)$ into $\pi_3(Fv, y) = M(v)$ by Corollary 39; the contraction (`orthog-contr`) is then fibrewise over Ab. Here the constancy used is that of $G : E \to \mathsf{Stage}[1]$ over the 1-connected total space E; this one step covers both their module M and their fibrewise $E_2(u)$, and working with $W : \mathsf{Stage}[1]$ directly avoids Corollary 39 and the 1-truncatedness of Ab; Ab enters only through Theorem 8 on the bases. The paper's TX is already large; Unit's Tor is small (over BAut), hence step 2. The base point is carried throughout: the point $F(v_0)$ comes from the pointing p of k through Theorem 36(1); nothing is recovered from connectedness at the end.

6. The Postnikov decomposition (Lemma 40)

Let $\| - \|_2 = \texttt{(Truncations.TO three)}$ ($\mathsf{isTrunc}\,4$ with $\mathsf{isConn}\,4$ fibres of the unit η).

Proposition 12. *$\alpha : \sum_{V:\mathsf{Stage}[0]} \mathsf{Fam}\,V \to \mathsf{SC3}$, $\alpha(V, F, f, c) = (\sum_v Fv, (v_0, f), \mathsf{isConn}, \mathsf{isTrunc})$, is an equivalence, with inverse $\delta(X, x, c, t) = (V_X,\ \mathrm{fib}_\eta,\ (x, \mathsf{refl}),\ \ldots)$, $V_X = (\|X\|_2, \eta x, \ldots)$.*

Proof. Levels: $\sum F$ is $\mathsf{isConn}\,3$ by Lemma 6 ($\mathsf{isConn}\,4$ fibres lowered by `conn-down`) and $\mathsf{isTrunc}\,5$ by `trunc-Sigma` ($V.X$ raised from $\mathsf{isTrunc}\,4$ by `trunc-suc`); $\|X\|_2$ is $\mathsf{isConn}\,3$ by `conn-up` along η (fibres $\mathsf{isConn}\,4$, lowered to $\mathsf{isConn}\,3$); $\mathrm{fib}_\eta(z)$ is $\mathsf{isConn}\,4$ (`TO.conn`) and $\mathsf{isTrunc}\,5$.

$\alpha\delta = \mathrm{id}$: $w \mapsto w.2.1 : \sum_z \mathrm{fib}_\eta z \simeq X$ (the text of `Presentation.Pres.tot-fib`, re-proved generically) sends $(\eta x, (x, \mathsf{refl}))$ to x by definition; Lemma 3.

$\delta\alpha = \mathrm{id}$: let $E = \sum F$, $\hat{e} : \|E\|_2 \to V.X$ the extension of pr_1 (`ConnectedMaps.Elim.ext`), $\beta_w : \hat{e}(\eta w) = w.1$ (`ext-beta`).

- $\hat{e}$ is an equivalence. Inverse: $g(v)$ is the value of the constant map $y \mapsto \eta(v, y) : Fv \to \|E\|_2$ (`ConnectedMaps.null`: Fv is $\mathsf{isConn}\,4$ and $\|E\|_2$ is $\mathsf{isTrunc}\,4$, so constants are an equivalence), with $g_\beta : g(v) = \eta(v, y)$. $\hat{e}g = \mathrm{id}$: at v, the map $y \mapsto \mathsf{ap}_{\hat{e}}(g_\beta) \cdot \beta_{(v,y)} : Fv \to (\hat{e}(gv) = v)$ into a $\mathsf{isTrunc}\,3 \subseteq \mathsf{isTrunc}\,4$ type is constant (`null` again). $g\hat{e} = \mathrm{id}$: `Elim.ext` along η into $z \mapsto g(\hat{e}z) = z$, given at ηw by $\mathsf{ap}_g(\beta_w) \cdot g_\beta$.
- $t_z : \mathrm{fib}_\eta(z) \simeq F(\hat{e}z)$ is the composite $\mathrm{fib}_\eta(z) \simeq \sum_w \hat{e}(\eta w) = \hat{e}z$ ($\mathsf{ap}_{\hat{e}}$, an equivalence by `ap-equiv`) $\simeq \sum_w w.1 = \hat{e}z$ (precomposition with β_w^{-1}) $\simeq F(\hat{e}z)$ (`fib-pr1`), so that, by definition,
$$t_z(w, q) = \mathrm{transport}_F\big(\beta_w^{-1} \cdot \mathsf{ap}_{\hat{e}}(q)\big)(w.2).$$
- Coherence, with $w_0 = (v_0, f)$: $t_{\eta w_0}(w_0, \mathsf{refl}) = \mathrm{transport}_F(\beta_{w_0}^{-1})(f)$ by definition (`t-pt`), and transporting this along the pointing $\beta_{w_0} : \hat{e}(\eta w_0) = v_0$ of $\hat{e}$ gives f (`subst-cancel`).

These data give the path by an SIP for $\sum_V \mathsf{Fam}V$ (`sip`): induction on the pointed equivalence (`StageId.J`, from the contractibility of $\sum_T S \simeq_* T$) reduces to the identity, where $F' = F$ by function extensionality and univalence from t (`fam-P`), the point by `ua-beta` and the coherence, and $\prod_v \mathsf{C}$ is a proposition (`fam-path`). □

7. Agreement with the realisation (Note 42)

Proposition 13. *For* $(A, B, k) : \mathsf{Triples}$, $\Phi(A, B, k) = \rho(A, B, k)$ *in* $\mathsf{SC3}$.

Proof. By the definitions of the steps, $\Phi(A, B, k) = (\sum_{x:K(A,2)} \mathsf{el}(kx).1,\ (\mathrm{pt}, \mathsf{tr}_{k\,\mathrm{pt}}(k_0^{-1})), \ldots)$ (Theorem 8 enters only by $A \mapsto K_2A$, $B \mapsto K_3B$, definitionally the same spaces). The map $(x, q) \mapsto (x, \mathsf{tr}_{kx}(q^{-1})) : \mathrm{Fib}k \to \sum_x \mathsf{el}(kx).1$ is an equivalence (`Fibres.sigma-equiv` of `sym-equiv` and `F.equiv`, as in `Presentation.Pres.tot`) and sends (pt, k_0) to the base point by definition (`Fib` uses $k(x) = \mathrm{pt}$; hence the inverse q^{-1}). Lemma 3. This is Hart–Milner's `NGrp-Sinh-coc-fib-`$\simeq$. □

8. Consequence: $\pi_3(S^2) = \mathbb{Z}$ through the classification

`MapFib.Final.rep-fib` (under the S^2 axiom) proves the marked universal property $s^* : [P, F]_* \simeq [S^2, F]_*$, $P = \mathrm{Fib}\kappa$, for every fibre $F = \mathrm{Fib}k$ of a triple. By Theorem 2, every $X : \mathsf{SC3}$ is $\rho(\mathsf{sec}\, X)$ up to the path $\mathsf{sec\text{-}eq}\, X : \rho(\mathsf{sec}\, X) = X$ in $\mathsf{SC3}$; transporting `rep-fib` at the triple $\mathsf{sec}\, X$ along it gives $s^* : [P, X]_* \simeq [S^2, X]_*$ for every pointed simply connected 3-type X (`Pi3Classified.Final.rep`). This is the interface `Pi3Sphere.Conditional.MapThm`, and `Pi3Sphere.Conditional.pi3-S2-T` (with the constructed 3-truncation of S^2) gives `Pi3Classified.Final.pi3-S2`: $\pi_3(S^2) \cong \mathbb{Z}$ as groups, from the S^2 axiom alone, without the special presentation of `Presentation.Pres`. No classification of maps is needed: the mapping theorem supplies that part.

9. Declarations (checked)

All in `sag/pi3-s2-postnikov`, inside the segment `Theory` (parameters: univalence, images, surjectivity of images).

`ClassAb`: `R`, `Props`, `rest-prop`, `mk`, `ab-eta`, `mk-eq`, `ab-path`	Lemma 7
`ClassBase.PtProp[P].I.equiv/path`, `StageId[n].I.equiv/path`, `StageId.J`	Lemma 3
`ClassBase.stage-gpd`	Lemma 4
`ClassBase.conn-lift`, `Const`, `const-contr`	Lemma 5
`ClassBase.conn-back`, `fib-pr1`, `total-conn`	Lemma 6
`ClassBase.raise`, `prop-of-contr`, `prop-equiv`, `pair-path`, `ap-equiv`, `subst-cancel`	generic
`ClassEM.O2ab`, `Rec2`, `K2`, `K3`, `pi2`, `pi3`, `k2-pi2`, `k3-pi3`, `Band2`, `Band3` (`fun-def`, `add-hom`), `pi2-k2`, `pi3-k3`, `k2-equiv`, `k3-equiv`	Theorem 8
`ClassFib.C`, `Fam`, `TorW[W]` (`tau`, `toL`, `ofL`, `toL-equiv`)	Prop. 9, steps 1–2
`ClassFib.Key[X,v0,cX,F]`: `to-C`, `Uniq.contr`, `s-prop`, `Exist`, `of-C`, `key`	Lemma 10
`ClassFib.Chain[V]`: `e01`, `e12`, `e23`, `e34`, `fib-equiv`	Prop. 9

`ClassPost.SC3`, `alpha`, `Dl.delta`, `alpha-delta`, `fam-path`, `sip`, `DA[u]` (`ehat`, `ehat-equiv`, `t`, `t-pt`, `coh`), `delta-alpha`, `post-equiv`	Def. 1, Prop. 12
`PostnikovClassification.Triples`, `realise`, `base`, `chain`	Def. 1, the chain
`PostnikovClassification.Agree`, `realise-agree`	Prop. 13
`PostnikovClassification.realise-equiv`, `classification`	Theorem 2
`Pi3Classified.Final`: `rep`, `map-thm`, `pi3-S2`	§9

The final statements are

```
realise-equiv  : P1.isEquiv Triples CP.SC3 realise
classification : P1.Equiv Triples CP.SC3 = (realise, realise-equiv)
```

in `U (l+1)`, with SC3 the type `ClassPost.SC3` (no alias is declared, rule 2). The inverse and both inverse laws are the fields `sec`, `sec-eq`, `ret`, `ret-eq` of `realise-equiv`.

10. Scope, checks and residual diagnostics

The equivalence transports identity types, hence equivalences of the classified objects; it is not a classification of noninvertible maps nor an equivalence of homotopy categories. Assumptions: only the Theory parameters; no sphere axiom, choice, resizing or new HIT. The new modules use parameter-free top-level declarations of sphere-named files (`SphereAxiom.O2`, `sigma-base`; `SphereUniversal.three`, `five`) and of `Presentation` (`O3`, `Rec`, `deloop-equiv`); none of the segments `SphereAxiom.Sphere`, `Presentation.Pres/Final`, `Sphere3*`, `MapFib`, `CupSquare` is used by the classification. Only `Pi3Classified` (§9) uses `MapFib.Final` and `CupSquare.kappa`, inside its segment `Final` whose parameters are the S^2 axiom.

Checks (28 October 2026): fresh `seg All.seg`: 4461 declarations, 0 failed, 2.23 s; server `segc sag/pi3-s2-postnikov all`: 89 modules, 0 failed, 1.96 s; uniform checker `segd -opens`: all 20 specifications (the 19 existing ones and `classification.opens`, 355 entries, including `Pi3Classified`) with no deviation. Every new declaration checks in at most 0.01 s. `classification.opens` was generated from `-opened` and read against this note module by module; it is not a line-by-line textual audit. One residual diagnostic: the `refl` lemma `ClassEM.Band2.fun-def` ("fun x = loop0($\Omega e_1\, x$) by definition") opens the band's internals (`EM1.band`, `TrivG.kb/tb`, `TorId.AutM.fa`, `Trunc.induction`) when comparing `(KK.e 0).1.1` with `EMSpace.loop0.1`; it is isolated in that one declaration, and `Band2.add-hom`, `Band3.add-hom` no longer open it.